HPE MIST ACCESS ASSURANCE

Product overview

HPE Mist Access Assurance is a cloud-based network access and control service that ensures zero trust, identity-based network access, and full stack policy and segmentation assignments with end-to-end user experience visibility for the self-driving network. The service delivers a suite of access control functionality with a flexible, yet simple, authorization policy framework for onboarding guest, IoT, BYOD, and corporate devices. Client connection is controlled based on user and device identities, regulating access for devices connecting to the network. Access Assurance also provides access control services for devices leveraging 802.1X authentication and MAC Address Bypass for non-802.1X allowlisted, wired IoT devices.

Key benefits

— No hardware or VM deployment need with HPE Mist infrastructure

— Built-in global High Availability avoiding multiple cluster deployments

— Automatic run-time feature updates, security and vulnerability fixes

— Simplified policy management

— Full-stack end-to-end visibility for eases debugging enabling self-driving network

— Built-in PKI infrastructure

— Powerful cloud-to-cloud integrations enables end-to-end workflows

Product description

HPE Mist Access Assurance is a microservices-based, cloud network access control (NAC) service that enables enterprises to easily enforce a zero trust security model for the self-driving network. Access Assurance solves many complexity challenges associated with traditional NAC offerings by:

— Removing on-premises server hardware
— Inherently highly available and resilient services
— Enabling automatic at-run-time feature updates, security, and vulnerability fixes
— Fully packaged PKI infra eliminating specialized software overhead
— Full-stack visibility—Client-to-Cloud

Access Assurance simplifies onboarding of wired and wireless, managed or headless IoT devices with 802.1X authentication or MAC Authentication Bypass (MAB) methods. Built-in PKI infrastructure eliminates the need for yet another tool making it as a truly integrated solution.

Access Assurance supports native fingerprinting without the need of any sensors by analyzing the packets passing through the HPE Mist infrastructure. Knowing the identity of the devices connected to the network is the foundation for zero trust.

Access Assurance uses hundreds of different vectors to match the identity of the user and device, such as X.509 certificate attributes, user group memberships, device compliance and posture metrics, and location context. These vectors help determine identity-based network admission criteria, such as the network segment or microsegment a device should connect to and the network policy that should be dynamically applied to a user.

Access Assurance has built-in controls for onboarding and management of corporate devices with 802.1X authentication and MAC-less onboarding of non-802.1X IoT and BYOD devices. Access Assurance capabilities simplify IT operations and secure connections for headless IoT and BYOD devices via a Multiple Pre-Shared Key (MPSK) mechanism. It incorporates a full suite of access control functionality leveraging MPSK or Private Pre-Shared Key (PPSK) as a new type of identity and policy vector.

Devices can be onboarded using PSK portal, enabling onboarding workflows by automating PSK generation based on user identity, leveraging Security Assertion Markup Language (SAML) for an SSO experience. It enables seamless client device onboarding via mobile QR code or by typing a personalized passphrase without installing any client software.

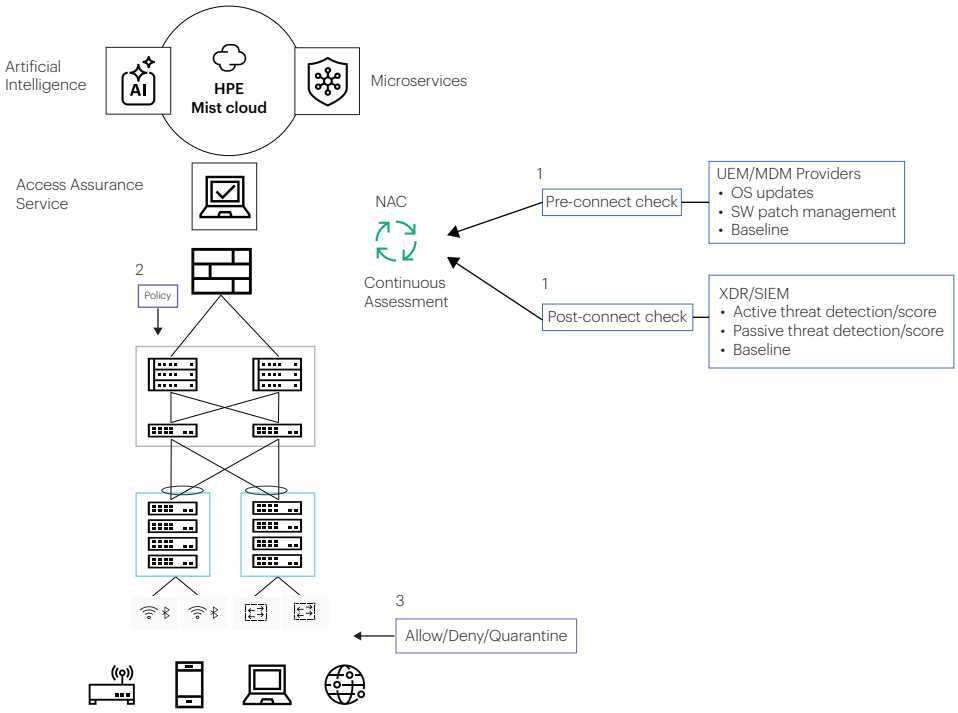


Figure 1. The HPE Mist Access Assurance cloud service greatly simplifies network access control

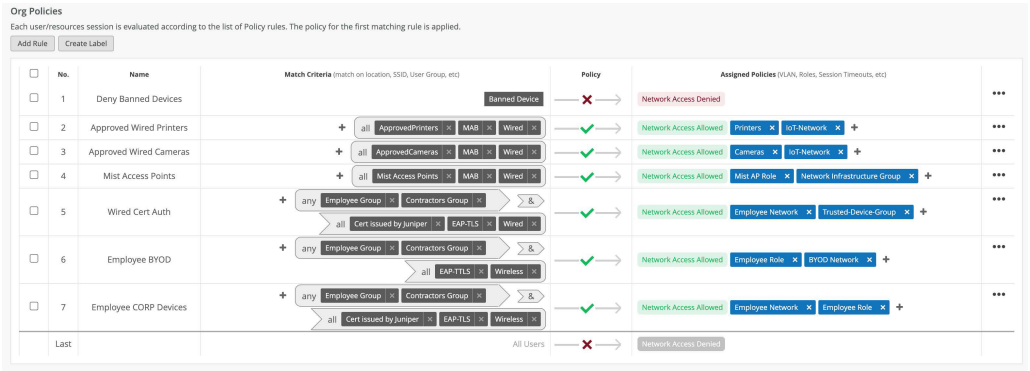


Figure 2. The flexible policy creation interface helps admins assign policies based on business requirements

Most importantly, Access Assurance provides end-to-end connectivity troubleshooting in a unified view from the client, network infrastructure, and access control perspective, dramatically simplifying day 2 support. IT admins gain a cohesive view of the end-user experience and can determine whether poor experiences are due to client configuration, network infrastructure, authentication, or a service.

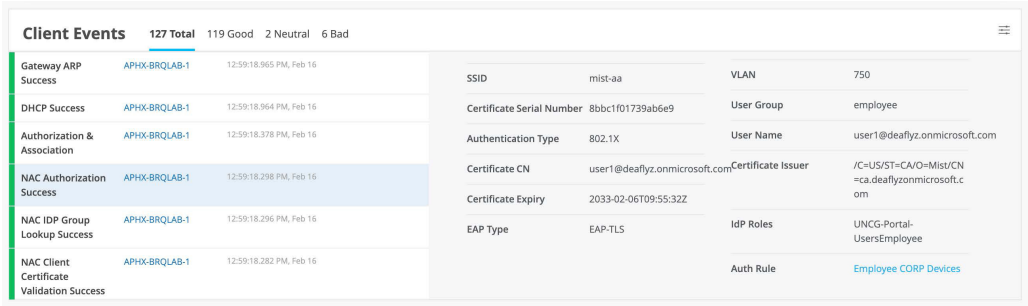


Figure 3. Client SLE tracks network access control events

Architecture and key components

Access Assurance is delivered through HPE Mist cloud and powered by HPE Mist AI. The microservices architecture ties together high availability, redundancy, and autoscaling for optimal network access across wired, Wi-Fi, and wide area networks. Using geo-awareness, Access Assurance automatically redirects authentication requests from different regions to the nearest Access Assurance instance to provide minimal latency and the best end-user experience.

Access Assurance provides an authentication service by integrating external directory services, like Google Workspace™, Microsoft Azure AD, Okta Identity, and others. It also integrates with Mobile Device Management (MDM) providers, such as Jamf, Microsoft Intune, and others, to provide granular user and device identification to enforce identity-based, zero trust network access control. Access Assurance has a built-in PKI infrastructure using which certificates can be easily distributed eliminating investment on additional tools.

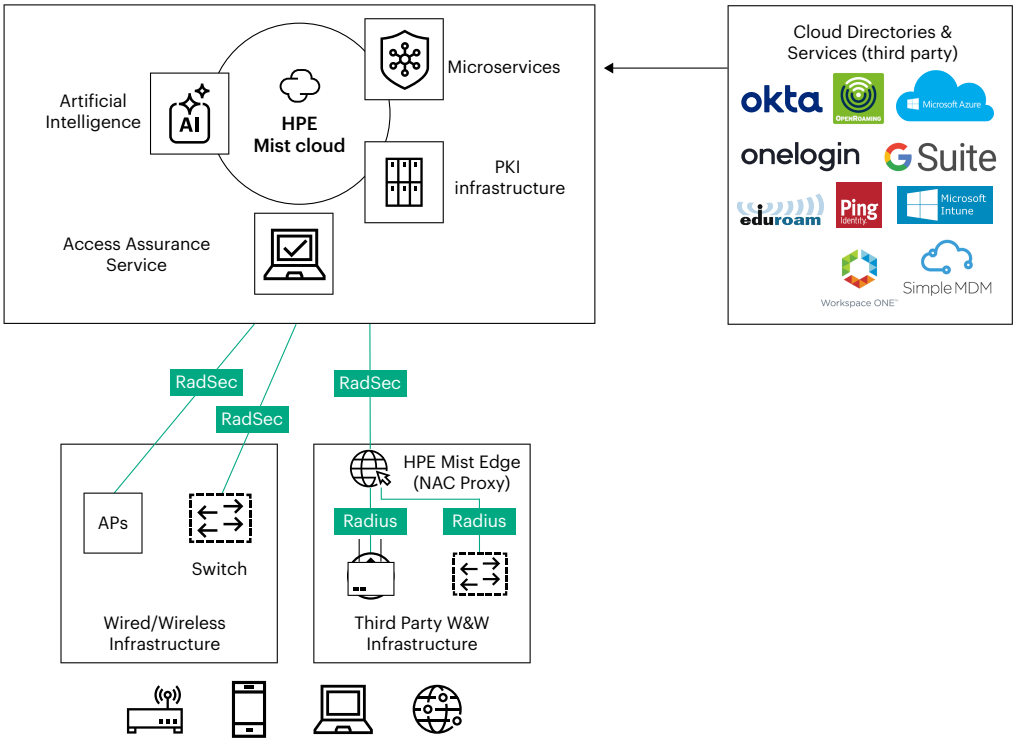


Figure 4. The HPE Mist cloud service greatly simplifies network access control

Features and benefits

Single pane of glass for management and operations

Access Assurance is tightly integrated with HPE Mist cloud, providing full-stack management and day-to-day operations for Wi-Fi Assurance, Wired Assurance, SD-WAN Assurance, and Access Assurance in one dashboard for end-to-end visibility. The Marvis AI Assistant engine leverages data from multiple sources for anomaly detection to provide actionable metrics. Through the dashboard, users can:

— Create and apply access policies that ensure only authorized devices and users are allowed network access

— Assign users and devices to the correct network segment

— Prevent users and devices from accessing restricted resources

— Add and modify certificates and certificate authorities

— Configure identity providers

— Monitor client activity across the stack

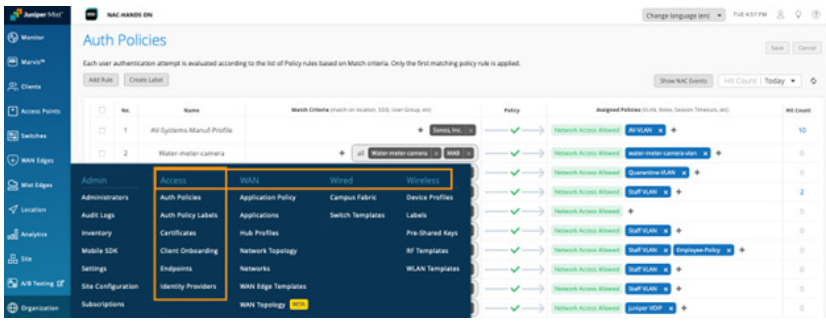


Figure 5. A friendly user interface highlights Access controls

Prioritizing client experiences

Access Assurance provides a unified view of the client connectivity experience and can easily identify a problem and perform root cause analysis. All client events, including connection and authentication successes and failures, are captured by HPE Mist cloud. With this data, the HPE Mist cloud helps simplify day-to-day operations by easily identifying if an end-user connectivity issue is caused by a client configuration mistake, network infrastructure and service problems, or authentication policy configuration issues. The HPE Mist service level expectations (SLEs) for wired and wireless clients are enhanced to include network access events, such as authentication events, certificate validations, and more.

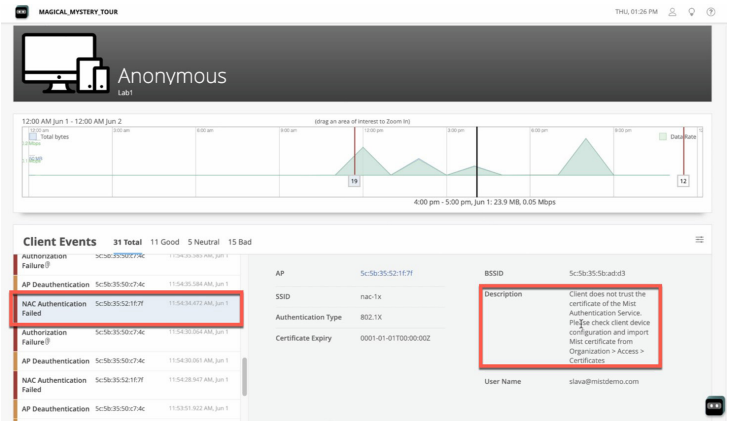


Figure 6. Client SLE failures provide descriptions for known issues

Granular user and device identity

Access Assurance can fingerprint user and device identities using various vectors without the need of external sensors, SPAN/TAP's or probes. Fingerprinting is done using various vectors like packet inspection, integrations and certificates.

The granular device and user identities can then be used for placing the devices and users in the appropriate part of network with appropriate access controls aligning with the organization's zero trust principles.

Access Assurance enables organizations to easily build their own connected devices CMDB which is helpful for compliance and enforce zero trust connectivity to the network.

New "Dry Run" capabilities within HPE Mist Access Assurance allow policies to be validated against actual conditions and to assess true impact before deployment, reducing risk, enabling zero trust, and ensuring operational continuity.

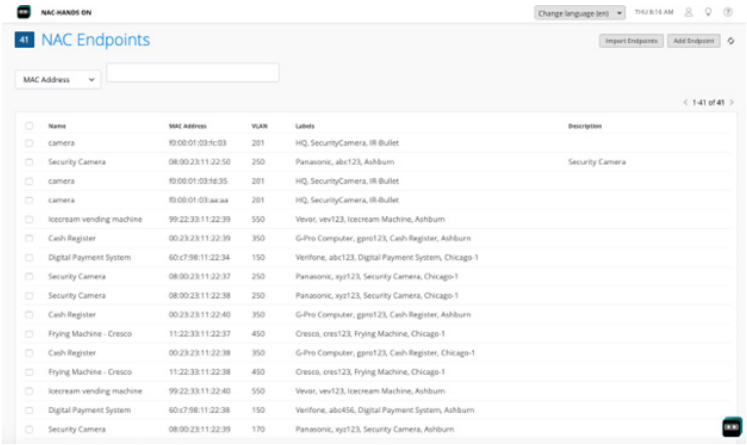


Figure 7. Identity fingerprinting is possible through multiple methods

OpenRoaming

OpenRoaming on HPE Mist enables users to connect automatically and securely to participating Wi-Fi networks using trusted identity credentials, creating a more seamless experience across locations. Built on standards-based technologies such as Passpoint and secure roaming frameworks, extend secure guest and public Wi-Fi access without repeated logins.

Network policy enforcement and microsegmentation

Based on user and device identity, Access Assurance can instruct the network to assign a user to a specific network segment (VLAN or a group-based policy tag), as well as enforce network policy by assigning a user role. Such roles can be leveraged in the HPE Mist WxLAN policy framework or switch policies.

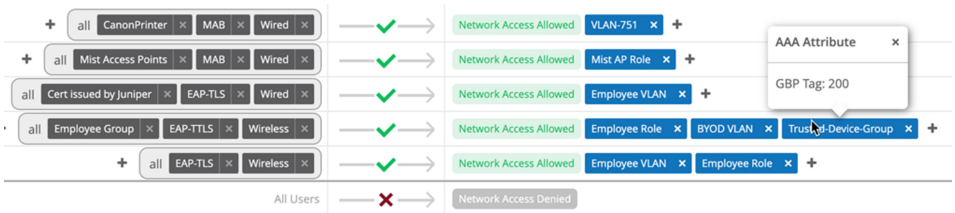


Figure 8. Enforced policies for VLANs, group-based policies, and user roles are easily visible

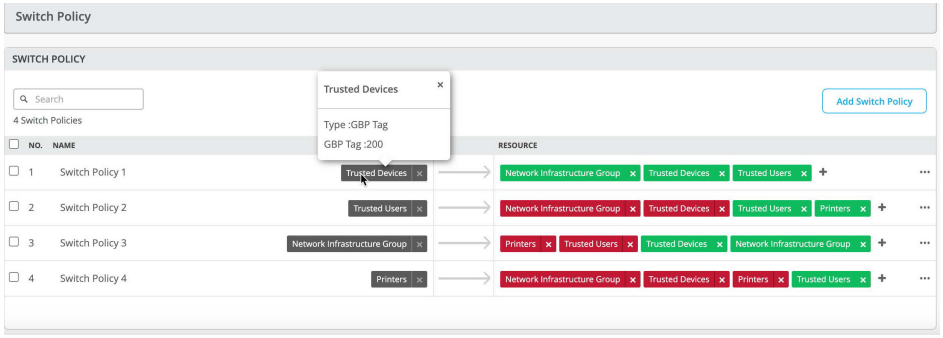


Figure 9. Recognizing policies with group-based tags is quick and fast

Built-in high availability and geo-affinity

With Access Assurance, organizations gain reliable and low-latency network access control of their networks in single and multisite deployments. HPE has deployed cloud instances of its network access control cloud service in multiple regional locations. In multisite deployments, authentication traffic coming from the network infrastructure is automatically directed to the nearest Access Assurance instance. Latency is minimized and users enjoy an exceptional wireless experience. This automated process is fully transparent to users and requires no involvement from the IT team. Organizations are assured reliable, redundant network access for client devices, regardless of the state of the nearest regional instance.

Automatic feature and security updates

HPE Mist microservices-based cloud architecture keeps Access Assurance optimized with the most advanced technologies. New features, security patches, and updates are automatically added to Access Assurance on a bi-weekly basis without interruptions or service downtime. This capability dramatically simplifies and improves service operations for network IT administrators, eliminating lengthy software upgrades and service downtime. HPE can easily deploy new features and functions to its cloud-based services, bringing advancements to market more rapidly and continuously improving your client-to-cloud experience.

HPE Marvis Virtual Network Assistant

HPE Marvis Virtual Network Assistant uses HPE Mist to help IT teams interact and engage with their networks. The Marvis AI engine binds together Access Assurance with other HPE Mist cloud-based services, such as HPE Mist Networking Wired Assurance, HPE Mist Networking Wi-Fi Assurance, and HPE Mist Networking WAN Assurance, helping the operations team move closer to achieving The Self-Driving Network™ with simplified troubleshooting and performance analysis.

Using features powered by HPE Mist, help desk staff and network administrators can simply ask a question in natural language and get actionable insights using the Marvis conversational interface that helps them identify and solve network issues. Marvis brings proactive anomaly detection into the SLE dashboard. With HPE Marvis Actions, staff gain proactive, actionable insights to identify network access issues across the full stack, providing recommendations for user connectivity issues. This provides our customers with easy root cause analysis across the full network stack and authentication services.

API-driven architecture

Access Assurance service is 100% based on public Representational State Transfer (REST) APIs that allow easy integration with external security information and event management (SIEM) or IT service management systems or other platforms for both configuration and policy assignment. These APIs provide the capability to invoke actions based on user or external events, as well as for using the cloud-native Webhook framework. Overall, the HPE Mist platform is 100% programmable, using OpenAPIs, for full automation and seamless integration with complementary HPE Networking access, wired, wireless, WAN, security, user engagement, and asset visibility solutions.

Specifications

Feature	Description
X.509 certificate management	— Built-in PKI infrastructure that can be used with MDM's or Marvis clients — External PKI support — Automatic CRL/OSCP certificate revocation check
External identity provider integration	The following protocols are supported to integrate into any identity provider to do user lookup and get device state information: — Secure Lightweight Directory Access Protocol (LDAP) — OAuth2 — eduroam secure network access — Integrations continually added for major unified endpoint management (UEM), enterprise mobility management (EMM) and mobile device management (MDM) tools
802.1X Authentication Methods	The following EAP methods are supported for secured 802.1X access: — Extensible Authentication Protocol-Transport Layer Security (EAP-TLS) — Protected Extensible Authentication Protocol PEAP TLS Tunnel — Extensible Authentication Protocol (TEAP) (TLS/TLS) — Extensible Authentication Protocol-Tunneled TLS (EAPTTLS)(PAP)
Non-802.1X authentication methods	MAC Authentication Bypass (MAB) including in-built CMDB and device fingerprinting Multi Pre-Shared Key (MPSK)
Network policy and microsegmentation	Assign VLANs, role and group-based policy tags dynamically based on the user and device identity
Third-party network infrastructure support	Supported via HPE Mist Edge Auth Proxy application, third-party vendor devices can communicate over standard RADIUS to the HPE Mist Edge Auth Proxy
Onboarding IoT devices	IoT and BYOD client-device onboarding — Create, rotate, auto-expire PSKs and MPSKs — Dynamic traffic engineering — Key-based WxLAN policy — Personal WLAN creation and management — Active device usage tracking per PSK — Automated key provisioning and rotation

Ordering information

The Access Assurance service is provided as a subscription, based on the average concurrently active client devices seen over a 7-day period.

Standard subscriptions include all basic network access control capabilities and IdP integration.

Advanced subscriptions add client posture checking (UEM/EMM/ MDM), Client-Onboarding using in-built PKI infra and basic fingerprinting to the standard Access Assurance capabilities.

SKU	Description
S-CLIENT-S-1	Standard Access & IoT Assurance subscription for 1 client for 1 year
S-CLIENT-S-3	Standard Access & IoT Assurance subscription for 1 client for 3 years
S-CLIENT-S-5	Standard Access & IoT Assurance subscription for 1 client for 5 years
S-CLIENT-S-7	Standard Access & IoT Assurance subscription for 1 client for 7 years
S-CLIENT-A-1	Advanced Access & IoT Assurance subscription for 1 client for 1 year
S-CLIENT-A-3	Advanced Access & IoT Assurance subscription for 1 client for 3 years
S-CLIENT-A-5	Advanced Access & IoT Assurance subscription for 1 client for 5 years
S-CLIENT-A-7	Advanced Access & IoT Assurance subscription for 1 client for 7 years
S-CLIENT-SS-M/1/3/5/7	Site Survivability licenses Monthly/1/3/5/7 years for 1 client subscription
S-CLIENT-SA-M	Standard to Advanced upgrade subscription for 1 client for 1 month

About HPE

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